

Syllabus for CS460 Computer Graphics VIS2026 Edition!

Web-based 3D Visualizations and beyond!

After completion of this course, you will be able to develop rich and interactive web-based 3D visualizations for computers, smartphones, and tablets! By using WebGL, you will learn to create immersive and high-quality graphics including 3D geometric transformations, camera transformations, the image-rendering process, as well as materials and texture mapping. This course also includes advanced topics such as color representations, light simulation, dealing with geometries, and creating animations. And since everything runs in the browser, you can show your work to your friends with a link to a website—how cool is that!

We will learn and use:

- JavaScript (including some HTML and CSS)
- XTK (<https://goXTK.com>)
- Three.js (<https://threejs.org>) and other computer graphics frameworks of your choice
- GLSL and "vanilla" WebGL and WebGPU!
- glTF (<https://www.khronos.org/glTF/>)
- Github / git (for version control and assignment submission)

Teaching Staff

Instructor: Daniel Haehn (he/him/his)

Teaching Assistants: Avani Kanamarlapudi (he/him/his)

Contact: @staff on Discord

Lectures

Tuesdays and Thursdays
12:30-1:45pm
University Hall Y01-1350 TEAL

Office Hours

Tuesdays and Thursdays
11:00am-12:30pm and by request
McCormack M03-2120, please use <https://calendly.com/haehn/> to reserve a slot.

Canvas Access

Please use Canvas to access course materials.
Login at <https://umassboston.instructure.com/>.

Discord Access

The teaching staff is available via Discord. The invite link will be shared in class.

Questions and Concerns

Please direct questions and concerns of any kind (now and during the semester) to the teaching staff in person or via Discord @staff or in the #help channel.

Course Conduct

Everybody is welcome here! This is a safe space! I strongly believe that the future will be inclusive but we need to be respectful towards each other even if opinions differ!

About the instructor and this course..

I am looking forward to meet you all and I really care about the success of every single one of you! Please know that I was struggling a lot in college. Especially with mathematics (I will tell you all about it in class). So this course will be very hands-on and not theoretical! I hope this helps most of you even though things can still be hard! Please don't hesitate to ask for help.

The TA Resource.

Besides myself, the TA is also getting paid to help you! Avanith will offer office/lab hours that will be announced in class.

Course Structure

27 Lectures

10 Assignments (30% of final grade, plus 10% bonus)

13 Quizzes (20% of final grade, take-home through Canvas)

Final Project (40% of final grade)

Participation (in-class, in-office, and as part of Discord discussions, 10% of final grade)

No exams

This semester, we will include more hands-on activities in class since the TEALS room supports it well.

Final Grade

The weighted scores from above will result in a final grade as follows:

$\geq 90 = A$	69-66 = C
89-86 = A-	65-62 = C-
85-82 = B+	61-58 = D+
81-78 = B	57-54 = D
77-74 = B-	53-50 = D-
73-70 = C+	below 50 = F

IEEE Visualization Visit

The IEEE Visualization conference is in Boston this year. We will do a Field Trip to visit the conference either 11/9, 11/10, or 11/12. I will cover the registration fee for everyone. This should be a great time!

Interactive Lectures (Bring your Laptop!)

Lectures will include interactive components. Please bring your laptop. If you do not have a laptop, please contact the teaching staff via Discord.

Assignments

Weekly assignments include research questions and coding challenges. All assignments require a written report in $\text{\LaTeX}$. You will use a standard git workflow paired with an online form to submit your work. **Assignments are due Tuesdays at 11:59pm. No late submissions.**

Quizzes

This year, we will do quizzes in class. More details to follow.

Final Project

The final project is the capstone of this course and counts as 40% of your final grade. This is your chance to apply your newly learned skills. You can fully enjoy the freedom and come up with your project idea or you can ask the teaching staff for suggestions. Project teams are encouraged but working alone is also fine. In the Fast Forward lecture, all students will present their final project plans in 2 minutes or less. Then, all students will present progress towards their final delivery during the Final Project Presentations. **And finally, all projects, including reports are due on 12/17/2026. No late submissions.**

Participation

In-class participation and Discord activity count towards your grade. If class attendance drops below 50%, surprise tests may happen during lectures.

Collaboration Policy and AI Policy

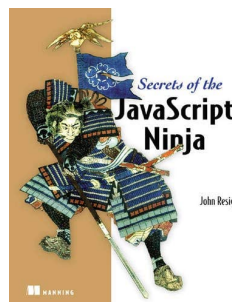
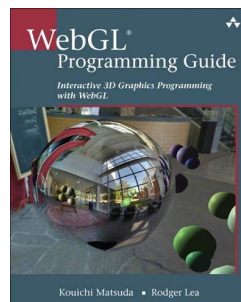
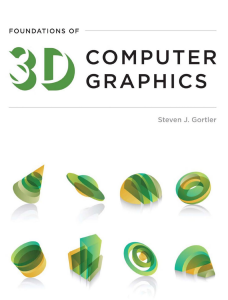
You are allowed and encouraged to collaborate with anybody. However, please make sure to give proper credit. For instance, if your friend helps you with an assignment or you copied code from another source, you must acknowledge their name in your code and the assignment report. **Students are expected to use AI tools and resources ethically and responsibly.** This includes ensuring that any data used in AI systems is properly licensed, and that any AI-generated content is appropriately cited.

Open Source License

The course material is publicly available under the MIT license (<https://opensource.org/licenses/MIT>). This includes assignment code. You are free to adopt a different license for your assignment solutions.

Readings

There are no required readings for this course. However, the material is based on the following three books:



- Foundations of 3D Computer Graphics by S. Gortler
- WebGL Programming Guide: Interactive 3D Graphics Programming with WebGL by K. Matsuda and R. Lea
- Secrets of the JavaScript Ninja by J. Resig and B. Bibeault

Limited copies of all books are available through the teaching staff. While the books are great, **you do not need to purchase them**—the most up-to-date information is available online.

Disability Accommodations

If you have a disability and feel you will need accommodation to complete course requirements, please contact the Ross Center for Disability Services at 617.287.7430.

Other Policies

We follow the Academic Policies of the Office of the Registrar.

See https://www.umb.edu/registrar/academic_policies or contact staff@cs460.org for questions.

Timeline

Date		Lecture	Due at 11:59pm
09/08/2026	Tu	01 Introduction and Web Developer Tools	
09/10/2026	Th	02 JavaScript and XTK Intro	Quiz 1
09/15/2026	Tu	03 Scene and Camera	Assignment 1 (TBA)
09/17/2026	Th	04 Three.js Intro	Quiz 2
09/22/2026	Tu	05 The Rendering Pipeline	Assignment 2 (TBA)
09/24/2026	Th	06 Shaders and WebGL I	Quiz 3
09/29/2026	Tu	07 WebGL II	Assignment 3 (TBA)
10/01/2026	Th	08 WebGL III	Quiz 4
10/06/2026	Tu	09 Transformations	
10/08/2026	Th	10 WebGPU (Avanith)	Quiz 5
10/13/2026	Tu	11 Quaternions and The Arcball	Assignment 4 (TBA)
10/15/2026	Th	12 Information Visualization	Quiz 6
10/20/2026	Tu	13 3D Picking	
10/22/2026	Th	14 Colors	Quiz 7
10/27/2026	Tu	15 Animations	Assignment 5 (TBA)
10/29/2026	Th	16 Generative AI	Quiz 8
11/03/2026	Tu	17 Textures	Assignment 6 (TBA)
11/05/2026	Th	18 Geometry	Quiz 9
11/10/2026	Tu	19 IEEE Visualization I	Assignment 7 (TBA)
11/12/2026	Th	20 IEEE Visualization II	Quiz 10
11/17/2026	Tu	21 Materials and Lighting	Assignment 8 (TBA)
11/19/2026	Th	22 Volume Rendering	Quiz 11
11/24/2026	Tu	23 glTF	Assignment 9 (TBA)
11/26/2026	Th	No Class (Thanksgiving Recess)	Quiz 12
12/01/2026	Tu	24 Final Project Fast Forward	
12/03/2026	Th	25 Outside Lecture	
12/08/2026	Tu	26 Skybox	Assignment 10 (TBA)
12/10/2026	Th	27 Final Project Presentations I	Quiz 13
12/15/2026	Tu	No class / Office hours only	
12/17/2026	Th	No class / Office hours only	Final Project